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DIAGNOSTIC SYSTEM, DIAGNOSTIC APPARATUS, AND DIAGNOSTIC METHOD

Abstract

A diagnostic system for a network includes the transmission node transmitting plural frames destined for the same reception node through the plural routes regardless of whether or not a failure has occurred in part of the plural routes the reception node performing measurement of respective arrival times of the plural frames received from the same transmission node on the basis of a timer and notifying the diagnostic apparatus of a result of the measurement and the diagnostic apparatus making, on the basis of the result of the measurement, a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the failure; and performance of the network after the reconstruction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and incorporates by reference the entire contents of Japanese Patent Application No. 2024-034404 filed in Japan on Mar. 6, 2024.

FIELD

[0002] The present invention relates to a diagnostic system, a diagnostic apparatus, and a diagnostic method.

BACKGROUND

[0003] Network routes are made redundant in preparation for device breakdowns in networks. For example, according to Japanese Laid-open Patent Publication No. 2018-045399 (hereinafter, Patent Literature 1), two Ethernet switches are connected by two lines, and normally, communication is executed by use of one of these lines. When a line malfunction detection circuit of the Ethernet switches detects a malfunction in the line being used, the route is switched by the line being changed to the other one of the lines.

[0004] In a technique disclosed in Patent Literature 1, a finite time period is required for the line malfunction detection circuit to detect the malfunction, a time period is required also for the route to be switched, and information transmission is thus interrupted.

[0005] An article (hereinafter, Non-Patent Literature 1) titled, “Industrial Ethernet Rings: PROFINet MRP and MRPD” (searched on Feb. 1, 2024 on the Internet, URL: https://www.hms-networks.com/news-and-insights/blog/posts/iot-blog/2018/06/11/industrial-ethernet-rings-profinet-mrp-and-mrpd), addresses the above described problem in Patent Literature 1. In a technique disclosed in Non-Patent Literature 1, communication routes are made redundant by routes having ring redundancy. Accordingly, in the technique disclosed in Non-Patent Literature 1, transmitting a communication frame to a destination through both routes in a ring prevents information transmission from being interrupted because even if one of the routes has a malfunction, the communication frame through the other route will arrive at the destination without delay.

[0006] The ring topology described in Non-Patent Literature 1 is difficult to be applied to a large-scale system with a large number of nodes because a propagation delay time period is generated at each node. Furthermore, topology having a mesh structure is often used in a standard Ethernet system and ring topology has less freedom in system construction.

[0007] Therefore, even in a case where topology having a mesh structure is used, a network diagnosis is desirably made so that information transmission is not interrupted.

[0008] For a network diagnosis, obtaining information from plural network switches enables checking of which route a communication frame is flowing through but requires a complicated process and is thus not practical.

[0009] Furthermore, in a case where a failure occurs somewhere in the network, a route for a communication frame is reconstructed in the mesh structure and the communication frame is passed through another route, but checking whether the mesh topology actually has enough redundancy for that reconstruction and getting a grasp of a time period lost for the reconstruction and influence by congestion of communication resulting from the reconstruction are very difficult.

[0010] In one aspect, an object is to provide a diagnostic system, a diagnostic apparatus, and a diagnostic method that enable efficient execution of diagnosis of a network having a mesh structure.

SUMMARY

[0011] It is an object of the present invention to at least partially solve the problems in the conventional technology.

[0012] According to an aspect of an embodiment, a diagnostic system for a network includes a

transmission node, a reception node, and a diagnostic apparatus, the network being configured to transmit a frame from the transmission node to the reception node by use of plural routes that are independent from one another other, the diagnostic system being configured to execute processing including: the transmission node transmitting plural frames destined for the same reception node through the plural routes regardless of whether or not a failure has occurred in part of the plural routes; the reception node performing measurement of respective arrival times of the plural frames received from the same transmission node on the basis of a timer and notifying the diagnostic apparatus of a result of the measurement; and the diagnostic apparatus making, on the basis of the result of the measurement, a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the failure; and performance of the network after the reconstruction.
[0013] The above and other objects, features, advantages and technical and industrial significance of this invention will be better understood by reading the following detailed description of presently preferred embodiments of the invention, when considered in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] FIG. **1** is a diagram illustrating a network having a double-tree structure;
[0015] FIG. **2** is a diagram illustrating a network having a mesh structure;
[0016] FIG. **3** is a diagram illustrating a diagnostic system according to an embodiment;
[0017] FIG. **4** is a diagram illustrating processing by a diagnostic apparatus;
[0018] FIG. **5** is a diagram illustrating an example of a data structure of a first measurement result table;
[0019] FIG. **6** is a diagram illustrating an example of a data structure of a second measurement result table;
[0020] FIG. **7** is a diagram illustrating an example of a data structure of a system vulnerability table;
[0021] FIG. **8** is a functional block diagram illustrating a configuration of the diagnostic apparatus according to the embodiment;
[0022] FIG. **9** is a flowchart illustrating a flow of processing by the diagnostic apparatus according to the embodiment;
[0023] FIG. **10** is a diagram illustrating an example of plural networks that are logically independent from one another; and
[0024] FIG. **11** is a diagram illustrating an example of a hardware configuration.
DESCRIPTION OF EMBODIMENTS
[0025] An embodiment of a diagnostic system, a diagnostic apparatus, and a diagnostic method disclosed by this application will hereinafter be described in detail on the basis of the drawings. The invention is not to be limited by this embodiment. Redundant description will be omitted as appropriate by assignment of the same reference sign to elements that are the same, and embodiments may be combined as appropriate so long as no contradiction arises from the combination.
EMBODIMENT
Network Having Mesh Structure
[0026] An example of a network relevant to a mesh structure will be described before description of an embodiment. FIG. **1** is a diagram illustrating a network having a double-tree structure. For example, a network N**1** has: distributed compute nodes (DCNs) **10**a, **10**b, **10**c, **10**d, and **10**e; DPX adapters **11**a, **11**b, **11**c, **11**d, and **11**e; HUBs **20** and **30**; and a setting unit **40**.

[0027] The DCNs **10***a* to **10***e* are respectively connected to the DPX adapters **11***a* to **11***e*. The DPX adapters **11***a* to **11***e* are each connected to the HUBs **20** and **30**. The DCNs **10***a* to **10***e* will each be generally referred to as the "DCN **10**" as appropriate. The DPX adapters **11***a* to **11***e* will each be generally referred to as a "DPX adapter **11**" as appropriate.
[0028] A DCN **10** outputs, data received from an input device and a control device, to a DPX adapter **11**.
[0029] Upon reception of the data from the DCN **10**, the DPX adapter **11** transmits communication frames through plural routes. Upon reception of the communication frames from the plural routes, a DPX adapter **11** selects any one of the communication frames and outputs data on the communication frame selected, to a DCN **10**.
[0030] The HUBs **20** and **30** are network switches. The network N**1** includes a communication route through the HUB **20** and a communication route through the HUB **30**. The setting unit **40** has been connected to the HUB **20**. The setting unit **40** performs various kinds of setup for the DCNs **10**, via the HUB **20**. Unlike communication between the DCNs **10**, communication between the setting unit **40** and the DCNs **10** does not require high reliability. For example, in a case where the HUB **20** has a failure, the setting unit **40** can be connected to the HUB **30** instead of the HUB **20**.
[0031] For example, upon reception of data from the DCN **10***b*, the DPX adapter **11***b* generates two communication frames that are identical to each other. The DPX adapter **11***b* transmits one of the communication frames to the DPX adapter **11***c* via the HUB **20** and the other communication frame to the DPX adapter **11***c* via the HUB **30**.
[0032] Upon reception of the two communication frames through the plural routes, the DPX adapter **11***c* selects any one of the communication frames and outputs data on the communication frame selected, to the DCN **10***c*.
[0033] The network N**1** described by reference to FIG. **1** has merits different from those of a ring configuration. For example, the merits include: (1) having more freedom and intuitiveness in its wiring than the ring configuration; (2) having no external disturbance upon insertion or elimination of a communication node; (3) having no risk of trouble being caused due to the ring being disconnected at two points at the same time; and (4) not requiring a ring master function needed for Media Redundancy Protocol (MRP). The case where two (double-tree) communication routes are available has been described by reference to FIG. **1**, but the number of the communication routes may be n where n is a natural number equal to or larger than 3. For example, for a triple-tree structure, HUBs can be replaced one by one without reduction in its reliability.
[0034] The HUBs **20** and **30** described by reference to FIG. **1**, may form a network having plural network switches in a mesh-like manner, as illustrated in FIG. **2**. FIG. **2** is a diagram illustrating a mesh-like network.
[0035] As illustrated in FIG. **2**, a network N**2** has DCNs **10***a* to **10***e* and DPX adapters **11***a* to **11***e*. The network N**2** has SWs **20-1**, **20-2**, **20-3**, **20-4**, **20-5**, and **20-6**. The network N**2** has SWs **30-1**, **30-2**, **30-3**, **30-4**, **30-5**, and **30-6**.
[0036] Description related to these DCNs **10***a* to **10***e* and DPX adapters **11***a* to **11***e* is the same as the description related to the DCNs **10***a* to **10***e* and DPX adapters **11***a* to **11***e* in FIG. **1**. The SWs **20-1** to **20-6** and the SWs **30-1** to **30-6** are, network switches (Ethernet switches). In FIG. **2**, the SW **20-1** to SW **20-6** and the SW **30-1** to SW **30-6** are illustrated completely symmetrically to each other but are not necessarily completely symmetrical to each other. The network N**2** may be configured to be an asymmetrical system using more SW **20** switches than SW **30** switches so as to transmit information via many stages.
[0037] For example, the DPX adapters **11***a* and **11***b* are each connected to the SWs **20-1** and **30-1**. The DPX adapters **11***c*, **11***d*, and **11***e* are each connected to the SWs **20-3** and **30-3**.
[0038] The SWs **20-1** to **20-6** implement data communication according to a predetermined communication protocol, to construct optimum route information beforehand. On the basis of the route information that has been constructed, the SWs **20-1** to **20-6** transmit communication frames.

For example, when the SW **20-1** receives a communication frame destined for the DPX adapter **11***c* from the DPX adapter **11***b*, the communication frame arrives at the DPX adapter **11***c* via the SWs **20-1**, **20-2**, and **20-3**.
[0039] In a case where a failure occurs in any of the SWs **20-1** to **20-6**, route information is reconstructed.
[0040] The SWs **30-1** to **30-6** implement data communication according to a predetermined communication protocol, to construct optimum route information beforehand. On the basis of the route information constructed, the SWs **30-1** to **30-6** transmit communication frames. For example, when the SW **30-1** receives a communication frame destined for the DPX adapter **11***c* from the DPX adapter **11***b*, the communication frame arrives at the DPX adapter **11***c* via the SWs **30-1**, **30-2**, and **30-3**.
[0041] In a case where a failure occurs in any of the SWs **30-1** to **30-6**, route information is reconstructed.
[0042] FIG. **2** illustrates the example including the SWs **20-1** to **20-6** and SWs **30-1** to **30-6**, but any other SW may be included.
[0043] The network having the double-tree structure and the network having the mesh structure have been described above.
Diagnostic System
[0044] An example of a diagnostic system according to the embodiment will be described next. FIG. **3** is a diagram illustrating the diagnostic system according to the embodiment. As illustrated in FIG. **3**, a diagnostic system S**1** has: DCNs **10***a* to **10***e*; DPX adapters **51***a*, **51***b*, **51***c*, **51***d*, and **51***e*; SWs **20-1** to **20-6** and **30-1** to **30-6**; and a diagnostic apparatus **100**.
[0045] The DCNs **10***a* to **10***e* are respectively connected to the DPX adapters **51***a* to **51***e*. For example, the DPX adapters **51***a* and **51***b* are each connected to the SWs **20-1** and **30-1**. The DPX adapters **51***c*, **51***d*, and **51***e* are each connected to the SWs **20-3** and **30-3**. The diagnostic apparatus **100** is connected to the SWs **20** or SWs **30**, or in some cases, to both the SWs **20** and the SWs **30**, although illustration of their connective relations is omitted in FIG. **3**.
[0046] A network constructed of the SWs **20-1** to **20-6** and a network constructed of the SW **30-1** to **30-6**, which will be described by reference to FIG. **3**, have been illustrated as physically independent networks. For example, a network of the diagnostic system S**1** is a double-tree network. The network constructed of the SWs **20-1** to **20-6** and the network constructed of the SWs **30-1** to **30-6** may be physically independent or logically independent as long as these networks are independent from each other.
[0047] The DCNs **10***a* to **10***e* will each be generally referred to as a "DCN **10**" as appropriate. The DPX adapters **51***a* to **51***e* will each be generally referred to as a "DPX adapter **51**" as appropriate.
[0048] The SWs **20-1** to **20-6** implement data communication according to a predetermined communication protocol, to construct optimum route information beforehand. On the basis of the route information that has been constructed, the SWs **20-1** to **20-6** transmit communication frames. For example, when the SW **20-1** receives a communication frame destined for the DPX adapter **51***c* from the DPX adapter **51***b*, the communication frame arrives at the DPX adapter **51***c* via the SWs **20-1**, **20-2**, and **20-3**.
[0049] In a case where a failure occurs in any of the SWs **20-1** to **20-6**, the SWs **20-1** to **20-6** implement data communication, to reconstruct route information.
[0050] The SWs **30-1** to **30-6** implement data communication according to a predetermined communication protocol, to construct optimum route information beforehand. On the basis of the route information that has been constructed, the SWs **30-1** to **30-6** transmit communication frames. For example, when the SW **30-1** receives a communication frame destined for the DPX adapter **51***c* from the DPX adapter **51***b*, the communication frame arrives at the DPX adapter **51***c* via the SWs **30-1**, **30-2**, and **30-3**.
[0051] In a case where a failure occurs in any of the SWs **30-1** to **30-6**, the SWs **30-1** to **30-6**

implement data communication, to reconstruct route information.

[0052] The DCNs **10** are communication nodes that require high data exchange reliability. For example, a DCN **10** outputs data received from an input device and a control device, to a DPX adapter **51**.

[0053] Upon reception of data from the DCN **10**, the DPX adapter **51** transmits identical communication frames through plural routes. In the example illustrated in FIG. **3**, the plural routes include a route constructed of the SW **20-1** to **20-6** and a route constructed of the SWs **30-1** to **30-6**. Furthermore, upon reception of the communication frames from the plural routes, a DPX adapter **51** selects any one of the communication frames and outputs data on the communication frame selected, to a DCN **10**.

[0054] In the following description, a communication frame that uses the route constructed of the SWs **20-1** to **20-6** will be referred to as a "first communication frame". A communication frame that uses the route constructed of the SWs **30-1** to **30-6** will be referred to as a "second communication frame". In a case where a first communication frame and a second communication frame are not particularly distinguished from each other, the first and second communication frames will each be referred to as a communication frame.

[0055] A DPX adapter **51** has a "timer" that measures a time difference between arrivals of communication frames received from plural routes. The DPX adapter **51** discloses (transmits) a result of measurement by the timer, to the diagnostic apparatus **100**.

[0056] The diagnostic apparatus **100** is an apparatus that executes network diagnosis for the diagnostic system S**1**. FIG. **4** is a diagram illustrating processing by a diagnostic apparatus. The diagnostic apparatus **100** selects a SW where a failure is to be caused to occur artificially, from the SWs **20-1** to **20-6** and the SWs **30-1** to **30-6**, and also selects a port in that SW, the port being where the failure is to be caused to occur.

[0057] FIG. **4** illustrates an example of a case where the SW **30-2** has been selected as the SW where a failure is to be caused to occur artificially. Furthermore, FIG. **4** illustrates, as an example, a case where the DPX adapter **51***b* transmits communication frames to the DPX adapter **51***c*.

[0058] For example, the diagnostic apparatus **100** executes processing of causing a failure to occur artificially, and processing of collecting arrival time difference information.

[0059] The following description is on the processing in which the diagnostic apparatus **100** causes a failure to occur artificially. For example, the diagnostic apparatus **100** transmits a control signal that causes a failure to occur artificially, to the SW **30-2**. Specifically, the diagnostic apparatus **100** disables a specific port in the SW **30-2**. A route between the SW **30-2** and the SW **30-3** is thereby blocked. Upon detection of route blocking, the SWs **30-1** to **30-6** reconstruct route information.

[0060] The following description is on the processing in which the diagnostic apparatus **100** collects arrival time difference information. In an example described hereinafter, the DPX adapter **51***b* transmits communication frames to the DPX adapter **51***c*, and the DPX adapter **51***c* discloses a measurement result to the diagnostic apparatus **100**.

[0061] For example, the DPX adapter **51***b* transmits communication frames to the SW **20-1** and the SW **30-1**. A DPX adapter **51** that transmits communication frames is an example of a "transmission node". In the following description, a DPX adapter that transmits communication frames will be referred to as a transmission node as appropriate.

[0062] The DPX adapter **51***c* receives a first communication frame and a second communication frame from the DPX adapter **51***b*, measures a time difference between a time that the first communication frame was received and a time that the second communication frame was received, and discloses a result of this measurement to the diagnostic apparatus **100**. The measurement result has, set therein, in addition to information on the time difference, information identifying the transmission node that has transmitted the communication frames and the reception node (the DPX adapter **51***c* itself). A DPX adapter **51** that receives a communication frame is an example of a "reception node". In the following description, a DPX adapter that receives a communication frame

will be referred to as a reception node as appropriate. In a case where the DPX adapter **51***c* does not receive one of the first communication frame and the second communication frame, the DPX adapter **51***c* sets a time-out in the measurement result.
[0063] Before causing an artificial failure (pseudo failure) to occur, the diagnostic apparatus **100** obtains a measurement result (a pre-failure setting arrival time difference) from the DPX adapter **51***c*, and registers relations between the transmission node (for example, the DPX adapter **51***b*), the reception node (the DPX adapter **51***c*), and the pre-failure setting arrival time difference, in a first measurement result table **141**. Similarly, every time the diagnostic apparatus **100** obtains a measurement result from another reception node, the diagnostic apparatus **100** repeatedly executes processing of registering relations between the transmission node, the reception node, and the pre-failure setting arrival time difference, in the first measurement result table **141**. For example, the pre-failure setting arrival time difference corresponds to a first time difference.
[0064] FIG. **5** is a diagram illustrating an example of a data structure of a first measurement result table. As illustrated in FIG. **5**, the first measurement result table **141** has pre-failure setting arrival time differences registered in relation to pairs of transmission nodes and reception nodes. For example, FIG. **5** illustrates that the pre-failure setting arrival time difference for the pair of the transmission node **51***a* (the same as the DPX adapter **51***a*) and the reception node **51***c* (the same as the DPX adapter **51***c*) is “TDac0”.
[0065] Subsequently, after causing an artificial failure to occur, the diagnostic apparatus **100** obtains a measurement result (an alternative route construction time period) from the DPX adapter **51***c*, and registers relations between the transmission node (for example, the DPX adapter **51***b*), the reception node (the DPX adapter **51***c*), and the alternative route construction time period, in a second measurement result table **142**. Similarly, the diagnostic apparatus **100** obtains a measurement result from another reception node and repeatedly executes processing of registering relations between the transmission node, the reception node, and the alternative route construction time period, in the second measurement result table **142**.
[0066] Furthermore, after the diagnostic apparatus **100** has caused the artificial failure to occur and route information has been reconstructed, the diagnostic apparatus **100** obtains a measurement result (a post-failure setting arrival time difference) from the DPX adapter **51***c*, and registers relations between the transmission node (for example, the DPX adapter **51***b*), the reception node (the DPX adapter **51***c*), and the post-failure setting arrival time difference, in the second measurement result table **142**. Similarly, the diagnostic apparatus **100** obtains a measurement result from another reception node and repeatedly executes processing of registering relations between the transmission node, the reception node, and the post-failure setting arrival time difference, in the second measurement result table **142**. For example, the post-failure setting arrival time difference corresponds to a second time difference.
[0067] FIG. **6** is a diagram illustrating an example of a data structure of a second measurement result table. As illustrated in FIG. **6**, the second measurement result table **142** has post-failure setting arrival time differences and alternative route construction time periods that have been registered in relation to pairs of transmission nodes and reception nodes. For example, FIG. **6** illustrates that an alternative route construction time period is “TRac” and a post-failure setting arrival time difference is “TDac1”, for the pair of the transmission node **51***a* (the same as the DPX adapter **51***a*) and the reception node **51***c* (the same as the DPX adapter **51***c*).
[0068] On the basis of the first measurement result table **141** and the second measurement result table **142**, the diagnostic apparatus **100** executes diagnostic processing, and registers a diagnostic result in a system vulnerability table **143**. For example, in a case where an alternative route construction time period is long (longer than a predetermined time period determined beforehand), the diagnostic apparatus **100** makes a diagnosis, concluding that “construction of an alternative route is impossible” or “the alternative route construction time period is too long”, when a failure occurs in the selected SW. More specifically, in a case where a time-out has been set in a

measurement result from a reception node, the diagnostic apparatus **100** makes a diagnosis concluding that construction of an alternative route is impossible, if the time-out persists, and makes a diagnosis concluding that the alternative route construction time period is too long in a case where the time-out duration is long (that is, the alternative route construction time period is long).

[0069] Furthermore, the diagnostic apparatus **100** makes a diagnosis concluding that the change in performance after the construction of the alternative route is too much in a case where the difference between the pre-failure setting arrival time difference and the post-failure setting arrival time difference is large (larger than a preset threshold). The diagnostic apparatus **100** registers the transmission node, the reception node, and the diagnostic result in the system vulnerability table **143** in association with one another. The diagnostic apparatus **100** thereafter undoes the artificial failure.

[0070] The diagnostic apparatus **100** repeatedly executes the above described processing by changing the SW and port where an artificial failure occurs.

[0071] FIG. **7** is a diagram illustrating an example of a data structure of a system vulnerability table. As illustrated in FIG. **7**, the system vulnerability table **143** has, comprehensively set therein, diagnostic results in cases where failures are caused to occur artificially at the respective SWs. For example, FIG. **7** illustrates that a diagnosis has been made, the diagnosis concluding that the alternative route construction time period is too long for the pair of the transmission node **51***a* (the same as the DPX adapter **51***a*) and the reception node **51***d* (the same as the DPX adapter **51***d*) in a case where a failure occurs in the SW **20-2** (Port 4). Furthermore, FIG. **7** illustrates that a diagnosis has been made, the diagnosis concluding that the change in the performance after construction of an alternative route is too much for the pair of the transmission node **51***a* (the same as the DPX adapter **51***a*) and the reception node **51***d* (the same as the DPX adapter **51***d*) in the case where a failure occurs in the SW **20-2** (Port 4).

Functional Configuration of Diagnostic Apparatus **100**

[0072] An example of a configuration of the diagnostic apparatus **100** according to the embodiment will be described next. FIG. **8** is a functional block diagram illustrating a configuration of the diagnostic apparatus according to the embodiment. As illustrated in FIG. **8**, the diagnostic apparatus **100** has a communication unit **110**, an input unit **120**, a display unit **130**, a storage unit **140**, and a control unit **150**. Functional units that the diagnostic apparatus **100** has are not limited to those illustrated in FIG. **8** and the diagnostic apparatus **100** may have any other functional unit. The diagnostic apparatus **100** may be implemented by plural server computers.

[0073] The communication unit **110** executes data communication with the DPX adapters **51***a* to **51***e* and the SWs **20-1** to **20-6** and **30-1** to **30-6**.

[0074] The input unit **120** inputs various kinds of information to the control unit **150** of the diagnostic apparatus **100**. The input unit **120** is, for example, a keyboard, a mouse, and a touch panel.

[0075] The display unit **130** displays information output from the control unit **150** of the diagnostic apparatus **100**.

[0076] The storage unit **140** stores the first measurement result table **141**, the second measurement result table **142**, and the system vulnerability table **143**. The storage unit **140** is implemented by, for example, a memory and a hard disk.

[0077] Description related to the first measurement result table **141**, the second measurement result table **142**, and the system vulnerability table **143** is the same as the description with respect to FIG. **5** and FIG. **6**.

[0078] The control unit **150** is a processing unit that governs the whole diagnostic apparatus **100**, and is implemented by a processor. This control unit **150** has a failure occurrence processing unit **151**, a collection unit **152**, and a diagnostic processing unit **153**.

[0079] The failure occurrence processing unit **151** selects a SW (and a port) where a failure is to be caused to occur artificially, and transmits a control signal that causes a failure to occur artificially, to the selected SW. After a diagnosis related to the SW where the failure has been caused to occur artificially is completed, the failure occurrence processing unit **151** undoes the artificial failure at the SW and changes the SW where a failure is to be caused to occur artificially.

[0080] The failure occurrence processing unit **151** may randomly select a SW where a failure is to be caused to occur artificially or may select the SW according to degrees of priority. For example, the degrees of priority of the SWs have been set beforehand.

[0081] Other processing executed by the failure occurrence processing unit **151** is the same as the processing, in which an artificial failure occurs and which has been described by reference to FIG. **4**.

[0082] Before an artificial failure occurs in a SW, the collection unit **152** obtains a measurement result (a pre-failure setting arrival time difference) from each reception node, and registers relations between the transmission nodes, the reception nodes, and the pre-failure setting arrival time differences, in the first measurement result table **141**.

[0083] After an artificial failure has occurred at a SW, the collection unit **152** obtains a measurement result (an alternative route construction time period) from each reception node and registers relations between the transmission nodes, the reception nodes, and the alternative route construction time periods, in the second measurement result table **142**.

[0084] After an artificial failure has been caused to occur and route information has been reconstructed, the collection unit **152** obtains a measurement result (a post-failure setting arrival time difference) from each reception node, and further registers relations between the transmission nodes, the reception nodes, and the post-failure setting arrival time differences, in the second measurement result table **142**.

[0085] On the basis of the first measurement result table **141** and the second measurement result table **142**, the diagnostic processing unit **153** executes diagnostic processing, and registers diagnostic results in the system vulnerability table **143**.

[0086] For example, the diagnostic processing unit **153** selects a pair of a transmission node and a reception node, and in a case where the alternative route construction time period is long (longer than a predetermined time period that has been determined beforehand), the diagnostic processing unit **153** makes, for the pair selected, a diagnosis concluding that construction of an alternative route is impossible or a time period required is too long if a failure occurs at the relevant SW (and port). For example, in a case where a time-out has been set in a measurement result, the diagnostic processing unit **153** makes a diagnosis concluding that construction of an alternative route is impossible and in a case where the alternative route construction time period is long, the diagnostic processing unit **153** makes a diagnosis concluding that the alternative route construction time period is too long.

[0087] The diagnostic processing unit **153** makes a diagnosis concluding that change in performance after construction of an alternative route is too much in a case where a difference between a pre-failure setting arrival time difference and a post-failure setting arrival time difference is large (larger than a preset threshold).

[0088] The collection unit **152** and the diagnostic processing unit **153** repeatedly execute the above described processing every time the SW (and port) where an artificial failure occurs is changed. The diagnostic processing unit **153** may output the diagnostic results registered in the system vulnerability table **143** to the display unit **130** and cause the display unit **130** to display the diagnostic results.

Flow of Processing

[0089] A flow of processing by the diagnostic apparatus **100** according to the embodiment will be described next. FIG. **9** is a flowchart illustrating a flow of processing by the diagnostic apparatus according to the embodiment. As illustrated in FIG. **9**, the failure occurrence processing unit **151** of the diagnostic apparatus **100** selects a SW where an artificial failure is to be caused to occur, from the SWs included in the communication routes (Step S**101**).

[0090] The collection unit **152** of the diagnostic apparatus **100** obtains a pre-failure setting arrival time difference before the artificial failure is caused to occur and registers the pre-failure setting arrival time difference in the first measurement result table **141** (Step S**102**). The failure occurrence processing unit **151** of the diagnostic apparatus **100** transmits a control signal for causing the artificial failure to occur, to the selected SW (Step S**103**).

[0091] The collection unit **152** obtains an alternative route construction time period and registers the alternative route construction time period in the second measurement result table **142** (Step S**104**). The collection unit **152** obtains a post-failure setting arrival time difference and registers the post-failure setting arrival time difference in the second measurement result table **142** (Step S**105**).

[0092] On the basis of the first measurement result table **141** and the second measurement result table **142**, the diagnostic processing unit **153** executes diagnostic processing and registers a diagnostic result in the system vulnerability table **143** (Step S**106**). The diagnostic processing executed by the diagnostic processing unit **153** at Step S**106** is the same as that described above.

[0093] The failure occurrence processing unit **151** undoes the failure at the SW where the artificial failure has been caused to occur (Step S**107**) and proceeds to Step S**101**.

Effects

[0094] Effects of the diagnostic system S**1** according to the embodiment will be described next. In the diagnostic system S**1**, a transmission node transmits plural communication frames destined for the same reception node through plural routes after an artificial failure has occurred in part of the plural routes. The reception node performs measurement of arrival times of the plural frames received from the same transmission node on the basis of a timer and notifies the diagnostic apparatus **100** of a result of this measurement. On the basis of the result of the measurement, the diagnostic apparatus **100** makes a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the artificial failure; and performance of the network after the reconstruction. Diagnosis of a network having a mesh structure is thereby able to be executed efficiently.

[0095] Furthermore, in the diagnostic system S**1**, the diagnostic apparatus **100** further executes processing of selecting part of nodes included in the plural routes and causing an artificial failure to occur at the selected part of the nodes (SW). Influence of a failure that occurs at each node is thereby able to be diagnosed easily.

[0096] Furthermore, the reception node in the diagnostic system S**1** notifies the diagnostic apparatus **100** of an alternative route construction time period that is a time period, over which part of the plural frames from the same transmission node does not arrive after the artificial failure occurs in the part of the plural routes. The diagnostic apparatus **100** makes a diagnosis of the alternative route construction time period that is a time period required for reconstruction of a communication route in the network upon occurrence of the artificial failure. A time period from the occurrence of the failure to the reconstruction of the communication route is thereby able to be determined easily.

[0097] Before the artificial failure occurs in the part of the plural routes, the reception node notifies the diagnostic apparatus **100** of a pre-failure setting arrival time difference that is a difference between arrival times of the plural frames received from the same transmission node, and notifies the diagnostic apparatus **100** of a post-failure setting arrival time difference that is a difference between arrival times of plural frames received from the same transmission node, after the artificial failure has occurred in the part of the plural routes and route information has been reconstructed. On the basis of the pre-failure setting arrival time difference and the post-failure setting arrival time difference, the diagnostic apparatus **100** makes a diagnosis of a difference between a time period for arrival of a frame in the network before the reconstruction and a time period for arrival of a frame in the network after the reconstruction. Performance after the reconstruction of the communication route is thereby able to be determined, the reconstruction resulting from the occurrence of the failure.

[0098] Furthermore, in the diagnostic system S**1**, in a case where the reception node has not received the frame from part of the plural routes, the reception node notifies the diagnostic apparatus **100** of information indicating that the frame has not been received from the part of the plural routes. The route that is no longer able to be used by the occurrence of the failure is thereby able to be identified. Setting of an artificial failure is comprehensively performed for the whole network system. According to the description thus far, this setting of an artificial failure is performed for one point at a time and after setting for one point is undone, setting for the next point is performed. However, multiple failures may be simulated. That is, the setting may be done simultaneously for more than one point. However, if failures are set simultaneously over two independent networks (in a case where there are two independent systems), transmission of a signal from a transmission node to a reception node may not be performed, and simultaneous setting for such plural points is thus to be avoided.

Other Network Configuration

[0099] The network constructed of the SWs **20-1** to **20-6** and the network constructed of the SWs **30-1** to **30-6** have been described as physically independent networks by reference to the example in FIG. **3** but as long as the network constructed of the SWs **20-1** to **20-6** and the network constructed of the SWs **30-1** to **30-6** are independent from each other, these networks may be physically independent from each other or logically independent from each other. The diagnostic system S**1** also performs transmission and reception of frames that do not meet the condition of being addressed to the same receiving node.

[0100] FIG. **10** is a diagram illustrating an example of plural networks that are logically independent from each other. In a network illustrated in FIG. **8**, the plural networks logically independent from each other are configured by use of the virtual LAN (VLAN) technology. For example, in one VLAN, a DPX adapter **11***b* and a SW **20-1** are connected to each other and in another VLAN, a DPX adapter **11***b* and a SW **30-1** are connected to each other. Furthermore, in the one VLAN, a DPX adapter **11***c* and a SW **20-3** are connected to each other and in the other VLAN, a DPX adapter **11***c* and a SW **30-3** are connected to each other. Using Multiple Spanning Tree Protocol (MSTP) prescribed by IEC/IEEE 60802 and network switches supporting MSTP enables implementation of the topology described by reference to FIG. **10**.

Hardware

[0101] An example of a hardware configuration of the diagnostic apparatus **100** will be described next. FIG. **11** is a diagram illustrating the example of the hardware configuration. As illustrated in FIG. **11**, the diagnostic apparatus **100** has a communication device **6***a*, a hard disk drive (HDD) **6***b*, a memory **6***c*, and a processor **6***d*. These units illustrated in FIG. **11** are connected to one another via a bus.

[0102] The communication device **6***a* implements communication with the DPX adapters **11**, the SWs **20-1** to **20-6**, and the SWs **30-1** to **30-6**, of the diagnostic system S**1**. The HDD **6***b* stores a DB and a program that causes the functions illustrated in FIG. **8** to operate.

[0103] The processor **6***d* operates processes that perform the functions described in FIG. **8** by reading programs from the HDD **6***b* or similar storage devices and deploying them to the memory **6***c*, thereby executing the same processing as each processing unit shown in FIG. **8** For example, this processing executes the same functions as the processing units that the diagnostic apparatus **100** has. Specifically, the processor **6***d* executes a process that executes the same processing as the failure occurrence processing unit **151**, the collection unit **152**, and the diagnostic processing unit **153**.

[0104] As described above, the diagnostic apparatus **100** operates as a diagnostic apparatus that executes an information provision method by reading and executing the program. Furthermore, the diagnostic apparatus **100** may implement the same functions as those according to the above described embodiment by reading the program from a recording medium by means of a medium

reading device and executing the program read. The program referred to herein is not limited to being executed by the diagnostic apparatus **100**. For example, the present invention may also be similarly applied to a case where another computer or a server executes the program, or a case where another computer and a server execute the program in corporation with each other.
[0105] The program may be distributed via a network, such as the Internet. Furthermore, the program may be executed by being recorded in a computer-readable recording medium, such as a hard disk, a flexible disk (FD), a CD-ROM, a magneto-optical (MO) disk, or a digital versatile disc (DVD), and being read from the recording medium by a computer.
Others
[0106] The following are some examples of a combination of technical features disclosed herein.
[0107] (1) A diagnostic system for a network comprising a transmission node, a reception node, and a diagnostic apparatus, the network being configured to transmit a frame from the transmission node to the reception node by use of plural routes that are independent from one another other, the diagnostic system being configured to execute processing including: [0108] the transmission node transmitting plural frames destined for the same reception node through the plural routes regardless of whether or not a failure has occurred in part of the plural routes; [0109] the reception node performing measurement of respective arrival times of the plural frames received from the same transmission node on the basis of a timer and notifying the diagnostic apparatus of a result of the measurement; and [0110] the diagnostic apparatus making, on the basis of the result of the measurement, a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the failure; and performance of the network after the reconstruction. [0111] (2) The diagnostic system according to (1), wherein the diagnostic apparatus further executes processing of selecting part of nodes included in the plural routes and causing an pseudo failure to occur at the selected part of the nodes. [0112] (3) The diagnostic system according to (1) or (2), wherein the reception node notifies the diagnostic apparatus of a time period over which part of the plural frames from the same transmission node does not arrive at the reception node, after the pseudo failure has occurred in part of the plural routes. [0113] (4) The diagnostic system according to (3), wherein the diagnostic apparatus makes a diagnosis of the a time period required for reconstruction of a communication route in the network upon occurrence of the pseudo failure. [0114] (5) The diagnostic system according to any one of (1) to (4), wherein the reception node further executes processing of notifying the diagnostic apparatus of information in a case where the reception node has not received the frame from part of the plural routes, the information indicating that the reception node has not received the frame from the part of the plural routes. [0115] (6) The diagnostic system according to (3), wherein the reception node notifies the diagnostic apparatus of a first time difference that is a difference between arrival times of plural frames received from the same transmission node before occurrence of a pseudo failure in part of the plural routes, and notifies the diagnostic apparatus of a second time difference that is a difference between arrival times of plural frames received from the same transmission node after the occurrence of the pseudo failure in the part of the plural routes and reconstruction of route information. [0116] (7) The diagnostic system according to (6), wherein the diagnostic apparatus makes a diagnosis of a difference between a time period for arrival of a frame in the network before the reconstruction and a time period for arrival of a frame in the network after the reconstruction, on the basis of the first time difference and the second time difference. [0117] (8) A diagnostic apparatus, comprising: [0118] a memory; and [0119] a processor coupled to the memory and the processor configured to: [0120] collect, from a reception node that receives plural frames transmitted from the same transmission node, a result of measurement of arrival times, regardless of whether or not a failure has occurred in part of plural routes included in a network, the plural routes being independent from one another; and [0121] make, on the basis of the result of the measurement, a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the failure; and performance of the network after the

reconstruction. [0122] (9) A diagnostic method for a network comprising a transmission node, a reception node, and a diagnostic apparatus, the network being configured to transmit a frame from the transmission node to the reception node by use of plural routes that are independent from one another, the diagnostic method executing processing including: [0123] the transmission node transmitting plural frames destined for the same reception node through the plural routes regardless of whether or not a failure has occurred in part of the plural routes; [0124] the reception node performing measurement of respective arrival times of the plural frames received from the same transmission node on the basis of a timer and notifying the diagnostic apparatus of a result of the measurement; and [0125] the diagnostic apparatus making, on the basis of the result of the measurement, a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the failure; and performance of the network after the reconstruction.
[0126] According to an embodiment, diagnosis of a network having a mesh structure is able to be executed efficiently.
[0127] Although the invention has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A diagnostic system for a network comprising a transmission node, a reception node, and a diagnostic apparatus, the network being configured to transmit a frame from the transmission node to the reception node by use of plural routes that are independent from one another other, the diagnostic system being configured to execute processing including: the transmission node transmitting plural frames destined for the same reception node through the plural routes regardless of whether or not a failure has occurred in part of the plural routes; the reception node performing measurement of respective arrival times of the plural frames received from the same transmission node on the basis of a timer and notifying the diagnostic apparatus of a result of the measurement; and the diagnostic apparatus making, on the basis of the result of the measurement, a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the failure; and performance of the network after the reconstruction.
2. The diagnostic system according to claim 1, wherein the diagnostic apparatus further executes selecting part of nodes included in the plural routes and causing a pseudo failure to occur at the selected part of the nodes.
3. The diagnostic system according to claim 2, wherein the reception node notifies the diagnostic apparatus of a time period over which part of the plural frames from the same transmission node does not arrive at the reception node, after the pseudo failure has occurred in part of the plural routes.
4. The diagnostic system according to claim 3, wherein the diagnostic apparatus makes a diagnosis of a time period required for reconstruction of a communication route in the network upon occurrence of the pseudo failure.
5. The diagnostic system according to claim 1, wherein the reception node further executes notifying the diagnostic apparatus of information in a case where the reception node has not received the frame from part of the plural routes, the information indicating that the reception node has not received the frame from the part of the plural routes.
6. The diagnostic system according to claim 3, wherein the reception node notifies the diagnostic apparatus of a first time difference that is a difference between arrival times of plural frames received from the same transmission node before occurrence of an pseudo failure in part of the plural routes, and notifies the diagnostic apparatus of a second time difference that is a difference

between arrival times of plural frames received from the same transmission node after the occurrence of the pseudo failure in the part of the plural routes and reconstruction of route information.

7. The diagnostic system according to claim 6, wherein the diagnostic apparatus makes a diagnosis of a difference between a time period for arrival of a frame in the network before the reconstruction and a time period for arrival of a frame in the network after the reconstruction, on the basis of the first time difference and the second time difference.

8. A diagnostic apparatus, comprising: a memory; and a processor coupled to the memory and the processor configured to: collect, from a reception node that receives plural frames transmitted from the same transmission node, a result of measurement of arrival times, regardless of whether or not a failure has occurred in part of plural routes included in a network, the plural routes being independent from one another; and make, on the basis of the result of the measurement, a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the failure; and performance of the network after the reconstruction.

9. A diagnostic method for a network comprising a transmission node, a reception node, and a diagnostic apparatus, the network being configured to transmit a frame from the transmission node to the reception node by use of plural routes that are independent from one another, the diagnostic method executing processing including: the transmission node transmitting plural frames destined for the same reception node through the plural routes regardless of whether or not a failure has occurred in part of the plural routes; the reception node performing measurement of respective arrival times of the plural frames received from the same transmission node on the basis of a timer and notifying the diagnostic apparatus of a result of the measurement; and the diagnostic apparatus making, on the basis of the result of the measurement, a diagnosis of: a time period required for reconstruction of a communication route in the network upon occurrence of the failure; and performance of the network after the reconstruction.